ADEMP-PreReg for the Simulation Study "Statistical considerations regarding transportability of clinical prediction models"

Damian Brülhart[1], Simon Schwab[2,3], and Ulrike Held[1]

[1]Department of Biostatsistics at the Epidemiology, Biostatistics and Prevention Institute, University of Zurich, Zurich, Switzerland
[2]Research and Quality Management, Swisstransplant, Bern, Switzerland
[3]Center for Reproducible Science, University of Zurich, Zurich, Switzerland

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1 General Information

1.1 What is the title of the project?

Answer: Statistical considerations regarding transportability of clinical prediction models and application to prediction models in liver transplantation.

1.2 Who are the current and future project contributors?

Answer: Damian Brülhart, Ulrike Held, and Simon Schwab

1.3 Provide a description of the project.

Answer: We will investigate the performance of the UK DCD Risk Score for predicting the outcome one-year graft failure for liver transplantation under varying conditions. We will simulate data sets based on descriptive statistics of organ donors and recipients in Switzerland. The descriptives will be provided by Simon Schwab at Swisstransplant. The metrics used to evaluate the performance of the models will be the AUC, calibration intercept and the calibration slope.

1.4 Did any of the contributors already conduct related simulation studies on this specific question?

Answer: We did not conduct previous simulation studies on the performance of the aforementioned models.

2 Aims

2.1 What is the aim of the simulation study?

Answer: The aim of the study is to evaluate the transportability of clinical prediction models (namely the UK DCD Risk Score) and their robustness regarding differences between development and the target population.

3 Data-Generating Mechanism

3.1 How will the parameters for the data-generating mechanism (DGM) be specified?

Answer: The simulation will be parametric based on real data. The parameters will be extracted from descriptive statistics provided by Swisstransplant (means, standard deviations, correlations and prevalences). The parameters needed for prediction are donor age, donor BMI, functional donor warm ischaemia time, cold ischaemia time, recipient age, recipient Model for End-Stage Liver Disease (MELD) Score and an indicator for retransplantation.

Donor age and donor BMI are expected to be correlated and therefore modeled as the bivariate normal distribution

$$\begin{bmatrix} X_{\text{D.Age}} \\ X_{\text{D.BMI}} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_{\text{D.Age}} \\ \mu_{\text{D.BMI}} \end{bmatrix}, \begin{bmatrix} \sigma^2_{\text{D.Age}} & \rho\sqrt{\sigma^2_{\text{D.Age}}\sigma^2_{\text{D.BMI}}} \\ \rho\sqrt{\sigma^2_{\text{D.Age}}\sigma^2_{\text{D.BMI}}} & \sigma^2_{\text{D.BMI}} \end{bmatrix} \right), \tag{1}$$

where ρ is the correlation coefficient, σ^2 the variance and μ the mean. The means, standard deviations and the correlation used for random sampling will be estimated from the Swisstransplant summary statistics. In order to exclude impossible negative values in these two variables, donor age and BMI will both be truncated between $[0, \infty)$.
Functional donor warm ischaemia time (FWIT) will be sampled from the normal distribution

$$X_{\text{FWIT}} \sim \mathcal{N}(\mu_{\text{FWIT}}, \sigma^2_{\text{FWIT}}) \tag{2}$$

truncated between $[0, \infty)$ in order to exclude impossible negative values. The mean and standard deviation used for random sampling will be estimated from the summary statistics provided by Swisstransplant.
Cold ischaemia time is available from Swisstransplant as a dichotomized variable (> 6h versus ≤ 6h). It was also implemented as such in the UK DCD Risk Score. Therefore, we will sample from the Bernoulli distribution

$$X_{\text{CIT}>6} \sim \text{B}(p_{\text{CIT}>6}), \tag{3}$$

where $p_{\text{CIT}>6}$ is the probability of the cold ischaemia time being larger than six hours. This probability will be estimated from the Swisstransplant summary statistics as well.
Recipient age will be sampled from the normal distribution

$$X_{\text{R.Age}} \sim \mathcal{N}(\mu_{\text{R.Age}}, \sigma^2_{\text{R.Age}}) \tag{4}$$

truncated between $[0, \infty)$ in order to exclude impossible negative values. The mean and standard deviation used for random sampling will be estimated from the Swisstransplant summary statistics.
The recipient MELD is a score bound between $[6, 40]$ consisting of integers only. Therefore we will sample from the Poisson distribution

$$X_{\text{R.MELD}} \sim \text{Po}(\lambda_{\text{R.MELD}}) \tag{5}$$

truncated between $[6, 40]$. $\lambda_{\text{R.MELD}}$ will be estimated as the mean from the Swisstransplant summary statistics (method of moments estimator).
The retransplantation variable will be sampled from the Bernoulli distribution

$$X_{\text{retp}} \sim \text{B}(p_{\text{retp}}), \tag{6}$$

where p_{retp} is the probability of the recipient being up for retransplantation. This probability will also be estimated from the summary statistics provided by Swisstransplant.
The outcome will be simulated from the Bernoulli distribution

$$Y \sim \text{Be}(p_1) \tag{7}$$

with p_1 being the probability for one-year graft failure. In the case of logistic regression models, the probability will be estimated using the logistic regression coefficients

$$\hat{p}_1 = \frac{1}{1 + \exp(-\boldsymbol{\beta}\boldsymbol{X})}.$$

In case of the model used for simulation being a Cox proportional hazards model, the estimated survival is given by

$$S(t \mid \boldsymbol{X}) = S(t)^{\exp(\boldsymbol{\beta}\boldsymbol{X})}$$

(Steyerberg, 2020, p. 83) which can be transformed into the estimated one-year graft failure probability as

$$\hat{p}_1 = 1 - S(t = 1 \mid \boldsymbol{X}) = 1 - S(1)^{\exp(\boldsymbol{\beta}\boldsymbol{X})}.$$

The intercept in case of the logistic regression model (β_0) and the baseline one-year graft survival ($S(1)$) are estimated such that approximately 20% overall prevalence of one-year graft failure is reached. This deliberately matches the Swisstransplant data.

3.2 What will be the different factors of the data-generating mechanism?

Answer: We will vary the model used for the prediction of the one-year graft failure probability. Models used for this purpose include the UK DCD Risk Score model itself (Schlegel et al., 2018), a model developed at King's College (Khorsandi et al., 2017) and a model developed at UCLA (Hong et al., 2011). The regression coefficients – especially from Schlegel et al. (2018) – may also be varied to represent what happens when the influence of the predicting variables changes. Additionally, to explore the transportability, other factors such as sample size and population characteristics may be varied.

3.3 If possible, provide specific factor values for the DGM as well as additional simulation settings.

Answer: The logistic regression coefficients reported for the UK DCD Risk Score (Schlegel et al., 2018) seem implausible as log odds ratios (retransplantation odds ratio would be $\exp(8.571) \approx 5276$) and are therefore expected to be odds ratios, which can be seen in Table 1a. The hazard ratios reported for the UCLA model (Hong et al., 2011), which can be used for prediction with the Swisstransplant data are shown in Table 1b. The hazard ratios reported for the Kings College model (Khorsandi et al., 2017), which can be used for prediction with the Swisstransplant data are shown in Table 1c. Specific values for other varying conditions (e.g., sample size, population characteristics, ...) will be defined at a later timepoint.

3.4 If there is more than one factor: How will the factor levels be combined and how many simulation conditions will this create?

Answer: Multiple factors are of interest (e.g., choice of model and regression coefficients for outcome simulation, sample size and population characteristics). The conditions of

Table 1: Odds ratios (OR) and hazard ratios (HR) from the UK DCD (1a) (Schlegel et al., 2018), UCLA (1b) (Hong et al., 2011) and Kings College (1c) (Khorsandi et al., 2017) models used for one-year graft failure simulation.
D.age: donor age; D.BMI: donor body-mass-index; FWIT: functional donor warm ischaemia time; CIT: cold ischaemia time; R.age: recipient age; R.MELD: recipient model of end-stage liver disease score; retp: retransplantation

(a)

Parameter	OR
D.age > 60 y	1/0.084
D.BMI > 25	1/0.519
FWIT > 20 min	1/0.341
CIT > 6 h	1/0.791
R.age > 60 y	1/0.241
R.MELD > 25	1/0.109
retp TRUE	8.571

(b)

Parameter	HR
FWIT > 20 min	1.9
CIT > 6 h	2.7
retp TRUE	11.6

(c)

Parameter	HR
FWIT > 25 min	1.48
R.MELD > 25	2.75
retp TRUE	1.87

interest will be varied in a partially factorial manner, depending on whether an interaction between the conditions is to be expected and needs to be explored.

4 Estimands and Targets

4.1 What will be the estimands and/or targets of the simulation study?

Answer: The estimands of the simulation study will be the performance measures of the UK DCD Risk Score in the Swiss population and under different conditions. The performance measures used are the area under the receiver operating characteristic (ROC) curve (AUC), the calibration intercept, the calibration slope and the net benefit according to Vickers et al. (2016). The performance measures are described in more detail in Section 6.

5 Methods

5.1 How many and which methods will be included and which quantities will be extracted?

Answer: We will only use one measure – the UK DCD Risk Score – and compare its performance under different conditions. The Score lies between zero and twenty-seven. The point system for the UK DCD Risk Score can be seen in Table 2. For the presence of each factor, a certain amount of points will be awarded, thus resulting in a higher score.

Table 2: Point system for the calculation of the UK DCD Risk Score. Whenever a factor is present in the donor-recipient combination, the corresponding amount of points will be awarded (Schlegel et al., 2018).

Condition	Points
Donor age > 60	2
Donor BMI > 25	3
Functional donor warm ischaemia time $\in (20, 30]$ min	3
Functional donor warm ischaemia time > 30 min	6
Cold ischaemia time > 6 h	2
Recipient age > 60	3
Recipient lab MELD > 25	2
Retransplantation TRUE	9

6 Performance Measures

6.1 Which performance measures will be used?

Answer: The performance measures used to evaluate the model under different conditions are the AUC, calibration slope and the calibration intercept. The AUC will be calculated using the `pROC` package in R (Robin et al., 2011).
The calibration intercept and slope will be calculated in R (R Core Team, 2025) by fitting a linear model on the actual probabilities versus the predicted probabilities. The actual probabilities will be binned on the three groups of predicted scores proposed by Schlegel et al. (2018) (low-risk: 0-5; high risk: 6-10; futile: 11-27). The predicted scores will be mapped onto a probability scale bound in $[0, 1]$, where the midpoint of the group is divided by 27. This means the predicted probabilities are always 2.5/27, 7.5/27 and 18.5/27 and the actual probability is the prevalence of the outcome in each of the three groups of predicted probabilities. An example for a calibration curve with intercept 0.06 and slope 0.97 can be seen in Figure 1.
The net benefit will be calculated as

$$\frac{\text{True positives}}{N} - \frac{\text{False positives}}{N} \cdot \frac{p_t}{1 - p_t}$$

(Vickers et al., 2016), where the true positives is the number of transplantations without one-year graft failure, the false positives is the number of transplantations with one-year graft failure and p_t – the threshold probability – will be determined by transplantation experts as the minimum percentage of successful transplantations (without one-year graft failure) at which a surgeon would still transplant. E.g., if a surgeon carries out 100 transplantations and he/she would want at least 60 of these procedures to be successful (without one-year graft failure), the threshold probability would be 0.6 or 60%.

6.2 How will Monte Carlo uncertainty of the estimated performance measures be calculated and reported?

Answer: We will report the Monte Carlo standard error in tables and the 2.5% as well as the 97.5% quantiles in plots. The Monte Carlo standard error will be calculated

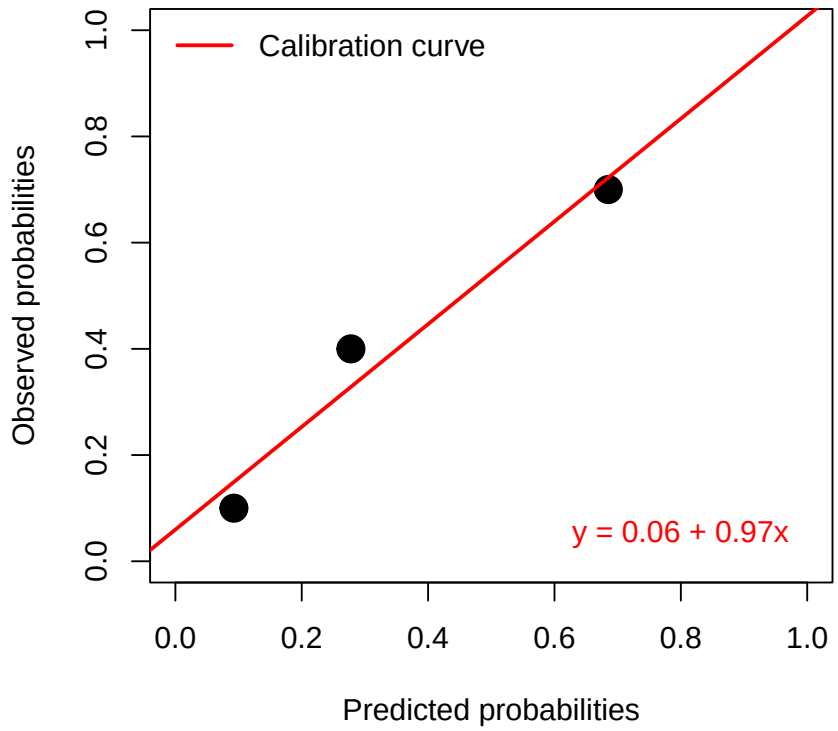


Figure 1: Example of a calibration curve. The linear model was fitted on the three points visible as black dots. The calibration intercept and slope can be seen as text in the plot (intercept: 0.06; slope: 0.97).

empirically as MCse $= \sqrt{\text{Var}(\phi)/n_{\text{sim}}}$ where ϕ is the corresponding performance measure (e.g., AUC) and n_{sim} is the number of simulation iterations (the number of times the performance measure is calculated).

6.3 How many simulation repetitions will be used for each condition?

Answer: The MCse will be reported for all performance measures. For the AUC, a MCse of 0.01 is desired. We will start with a simulation size of $n_{\text{sim}} = 1000$ iterations, which will be increased if the MCse is larger than desired.

6.4 How will missing values due to non-convergence or other reasons be handled?

Answer: Missing values and non-convergence are not expected. If any missing values or non-convergence are observed, we will handle them according to the proposed methods by Pawel et al. (2024). We will report the number and proportion of missing values and non-converged cases per simulation condition.

6.5 How do you plan on interpreting the performance measures? (optional)

Answer: The performance measures will be interpreted in an exploratory manner. They will be used to compare how the UK DCD risk score performs under the different conditions. An AUC of 1 describes a perfect model, whereas an AUC of 0.5 describes a model which is no better than chance and an AUC under 0.5 describes a harmful model under the underlying simulation conditions. We expect the perfect model to have a calibration intercept of 0 and a calibration slope of 1. The larger or smaller they are, respectively, the worse the calibration of the model under the underlying simulation condition. The simulation runs using the UK DCD Risk Score for outcome simulation as well as the prediction is expected to be the one with the best performance measures and therefore will be handled as the "perfect" or "most optimistic" model.

7 Other

7.1 Which statistical software/packages do you plan to use?

Answer: We will use the following packages of R version 4.5.0 (R Core Team, 2025) in their most recent versions: The `MASS` package (Venables & Ripley, 2002) to sample from the multivariate normal distribution using the `mvrnorm()` function, the `auc()` function of the `pROC` package (Robin et al., 2011) for calculation of the AUC, the `SimDesign` package (Chalmers & Adkins, 2020) to set up and run the simulation study and the `ggplot2` package (Wickham, 2016) for visualization. Additional packages may be utilized.

7.2 Which computational environment do you plan to use?

Answer: The simulation will be run on a MacOS Sequoia machine. The complete output of `sessionInfo()` will be made available online (e.g., GitHub) as well as in the supplementary material.

7.3 Which other steps will you undertake to make simulation results reproducible? (optional)

Answer: The entire code used in the simulation will be provided online (e.g., GitHub).

7.4 Is there anything else you want to preregister? (optional)

Answer: No.

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